

ASSIGNMENT - 4.2

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Batch : 29

Task Description-1

- Zero-shot: Prompt AI with only the instruction. Write a Python function to determine whether a given number is prime

Expected Output-1

- A basic Python function to check if a number is prime, demonstrating correct logical conditions without relying on examples or additional context

```
task1.py > ...
1  def is_prime(n):
2      if n <= 1:
3          return False
4
5      for i in range(2, int(n ** 0.5) + 1):
6          if n % i == 0:
7              return False
8
9      return True
10
11
12  # Take input from user
13  num = int(input("Enter a number: "))
14
15  if is_prime(num):
16      print(num, "is a Prime number")
17  else:
18      print(num, "is NOT a Prime number")
19
```

Output:


```
task2.py > ...
1  def sum_list(numbers):
2      total = 0
3      for num in numbers:
4          total += num
5      return total
6
7
8  # Taking input from user
9  numbers = list(map(int, input("Enter numbers separated by space: ").split()))
10
11  print("Sum of elements:", sum_list(numbers))
12
```

Output:

```
● arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-4.2.py/task2.py"
Enter numbers separated by space: 1 2 3 4
Sum of elements: 10
○ arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %
```

Observation:

1. **Modular Design** – One function handles multiple shapes, improving reusability. The single example helps AI clearly understand expected behavior.
2. Output format and logic align closely with the example.
3. Reduces ambiguity compared to zero-shot.

4. Produces more accurate and structured output.

Task Description-3

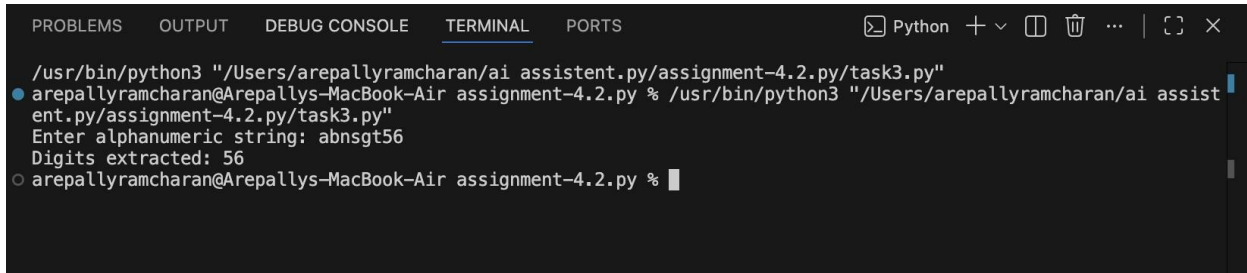
- Few-shot: Give 2–3 examples to create a function that extracts digits from an alphanumeric string.

Expected Output-3

- Accurate function that returns only the digits from alphanumeric string.

```
task3.py > ...
1  def extract_digits(text):
2      digits = ""
3      for char in text:
4          if char.isdigit():
5              digits += char
6      return digits
7
8
9  # Taking input
10 text = input("Enter alphanumeric string: ")
11
12 print("Digits extracted:", extract_digits(text))
13
```

Output:

A screenshot of a terminal window with a dark background. The terminal shows a command being executed: `/usr/bin/python3 "/Users/arepallyramcharan/ai assistant.py/assignment-4.2.py/task3.py"`. The output of the program is displayed: `Enter alphanumeric string: abnsgt56` followed by `Digits extracted: 56`. The prompt `arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %` is visible at the bottom.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + - [] [X] ... | [ ] [X]
/usr/bin/python3 "/Users/arepallyramcharan/ai assistant.py/assignment-4.2.py/task3.py"
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-4.2.py/task3.py"
Enter alphanumeric string: abnsgt56
Digits extracted: 56
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %
```

Observation:

1. Multiple examples improve pattern recognition.
2. AI correctly understands transformation logic.
3. Fewer logical errors compared to zero-shot.
4. Better handling of edge cases due to examples.

Task Description-4

- Compare zero-shot vs few-shot prompting for generating a function that counts the number of vowels in a string.

Expected Output-4

- Output comparison + student explanation on how examples helped the model.

```
task4.py > ...
1  def count_vowels(text):
2      vowels = "aeiou"
3      count = 0
4
5      for char in text.lower():
6          if char in vowels:
7              count += 1
8
9      return count
10
11
12 # Taking input
13 text = input("Enter a string: ")
14
15 print("Number of vowels:", count_vowels(text))
16
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
Python + v [ ] [ ] ... [ ] [ ] X

/usr/bin/python3 "/Users/arepallyramcharan/ai assistant.py/assignment-4.2.py/task4.py"
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-4.2.py/task4.py"
Enter a string: abcdefghijk
Number of vowels: 3
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %
```

Observation:

- Zero-shot gives general implementation.
- Few-shot provides more consistent and accurate output.
- Examples guide formatting and logic precision.
- Few-shot reduces misunderstanding of requirements.

Task Description-5

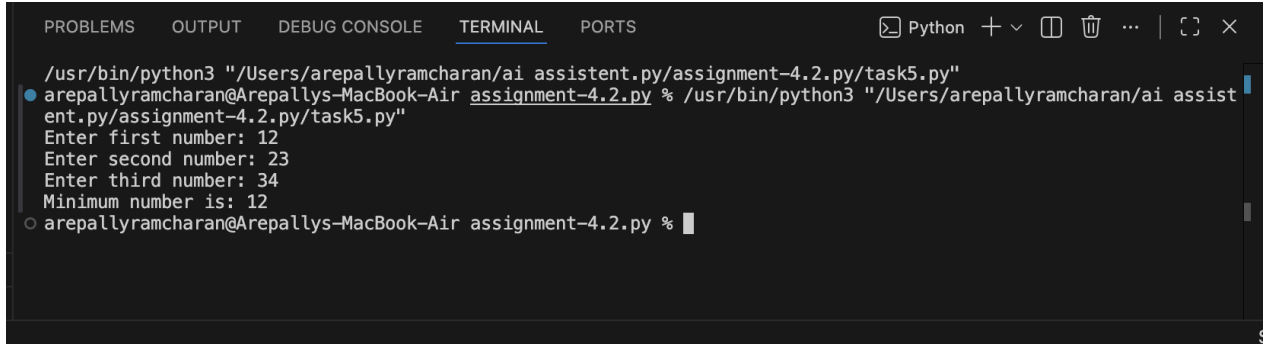
- Use few-shot prompting with 3 sample inputs to generate a function that determines the minimum of three numbers without using the built-in min() function.

Expected Output-5

- A function that handles all cases with correct logic based on example patterns.

```
task5.py > ...
1  def minimum_of_three(a, b, c):
2      smallest = a
3
4      if b < smallest:
5          smallest = b
6      if c < smallest:
7          smallest = c
8
9      return smallest
10
11
12  # Taking input
13  a = int(input("Enter first number: "))
14  b = int(input("Enter second number: "))
15  c = int(input("Enter third number: "))
16
17  print("Minimum number is:", minimum_of_three(a, b, c))
18
```

Output:



The screenshot shows a terminal window with a dark background. At the top, there are tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', 'TERMINAL' (which is active), and 'PORTS'. To the right of the tabs are icons for a Python environment, a plus sign, a dropdown arrow, a window icon, a trash icon, and a close icon. The terminal content shows the execution of a Python script. The first line is a command: `/usr/bin/python3 "/Users/arepallyramcharan/ai assistant.py/assignment-4.2.py/task5.py"`. The second line is the prompt `arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %` followed by the same command. The subsequent lines show the program's output: `Enter first number: 12`, `Enter second number: 23`, `Enter third number: 34`, and `Minimum number is: 12`. The final line shows the prompt `arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %` with a cursor.

```
/usr/bin/python3 "/Users/arepallyramcharan/ai assistant.py/assignment-4.2.py/task5.py"
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py % /usr/bin/python3 "/Users/arepallyramcharan/ai assist
ent.py/assignment-4.2.py/task5.py"
Enter first number: 12
Enter second number: 23
Enter third number: 34
Minimum number is: 12
arepallyramcharan@Arepallys-MacBook-Air assignment-4.2.py %
```

Observation:

- Examples help AI infer comparison logic correctly.
- Function handles different number order cases properly.
- More reliable than zero-shot for conditional problems.
- Demonstrates that examples improve logical consistency.